

Singular Value Decomposition

No-Square Matrix [rectangular]

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}_{(2 \times 2)} \Rightarrow \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}_{(2 \times 3)}$$

rows $\neq$ columns

Ex:- $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}_{2 \times 3}$ (3D)
 (2D) output Space input Space

input vector:- $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{(3 \times 1)}$ o/p:- $Ax = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

→ Dimension reduced from 3D → 2D

Rectangular Diagonal matrix:-

A matrix would be diagonal if it were square, but instead is rectangular due to extra rows or columns or zeroes

Ex:- $\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$
 rectangular diagonal matrix

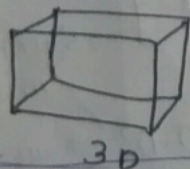
$$\begin{bmatrix} a & 0 \\ 0 & b \\ 0 & 0 \end{bmatrix}$$

2 Transformation

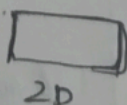
Ex:-1 $\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \end{bmatrix}_{2 \times 3} \Rightarrow \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{(2 \times 3)}$
 i) Stretching / scaling ii) dimension reduced from 3D to 2D

rectangular diagonal matrix $\Rightarrow$ linear

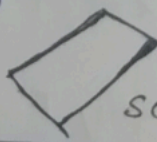
Transformation



3D



2D



scaling / stretching

Ex:- $\begin{bmatrix} a & 0 \\ 0 & b \\ 0 & 0 \end{bmatrix} \Rightarrow \left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}_{(3 \times 2)} \right\}$ matrix multiplying not possible

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}_{(3 \times 2)}$$

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}_{(2 \times 2)}$$

$$\Rightarrow (3 \times 2)$$

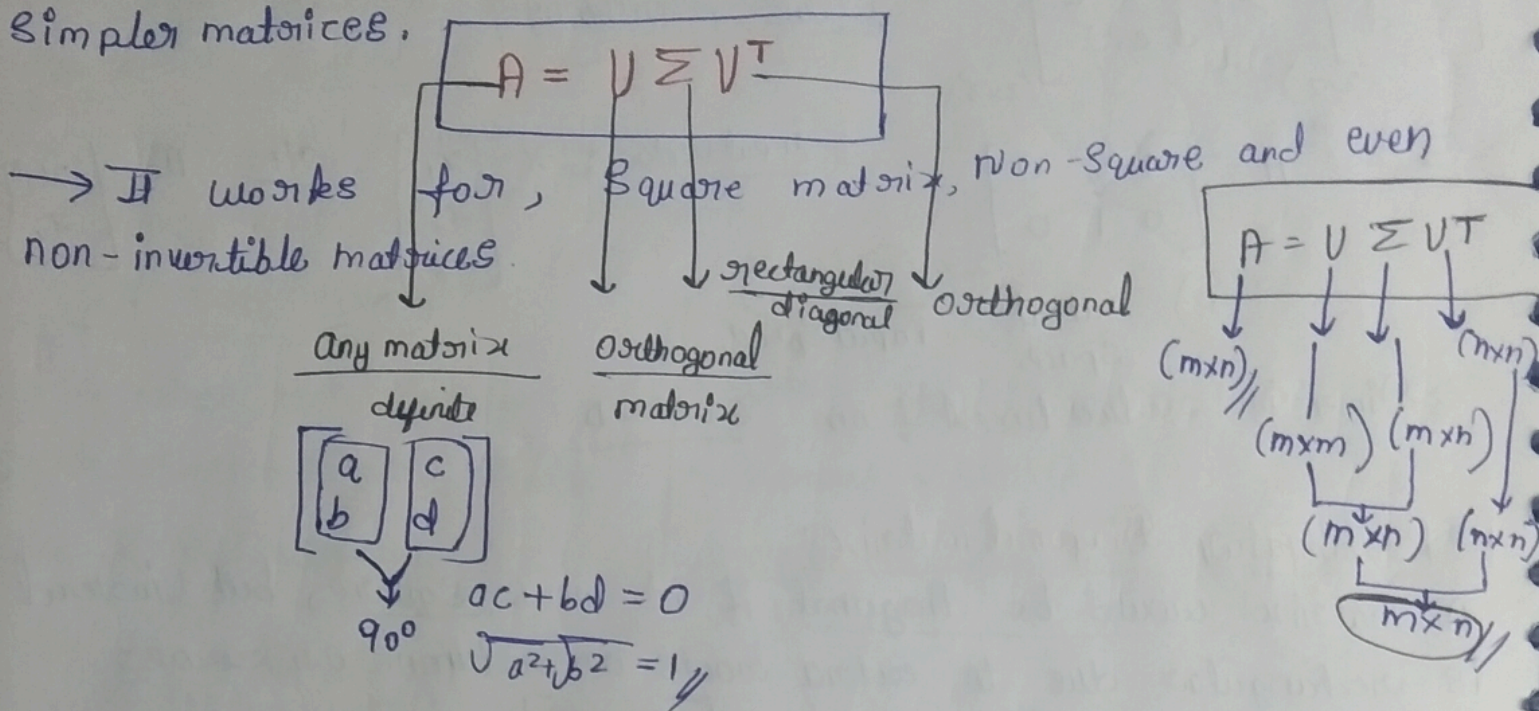
i) Stretching

ii) dimension increased from 2D → 3D

2D → 3D

SVD :-

SVD (Singular Value Decomposition) is a method to break any matrix (square or non-square) into three simpler matrices.



Relationship with Eigen Decomposition :-

$$A = U \Sigma V^T$$

A non square matrix

$$A = V \Lambda V^{-1}$$

A is symmetric

$$A \cdot A^T = U \Sigma V^T (U \Sigma V^T)^T$$

$$= U \Sigma V^T V \Sigma^T U^T$$

$$= U \Sigma \Sigma^T U^T$$

$$A \cdot A^T = U X U^T$$

where $X = \Sigma \Sigma^T$

$$A^T \cdot A = (U \Sigma V^T)^T U \Sigma V^T$$

$$= V \Sigma^T U^T U \Sigma V^T$$

$$= V \Sigma^T \Sigma V^T$$

$$A^T \cdot A = V Y V^T$$

where $Y = \Sigma^T \Sigma$

$\therefore V$ is orthogonal

$U \cdot V^T = I$

$V^T = V^T$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 1 & 1 \end{bmatrix}$$

2×3 3×2

$$A A^T = \begin{bmatrix} 6 & 1 \\ 1 & 1 \end{bmatrix}$$

⇒ Symmetric matrix

(If we apply $A \cdot A^T \Rightarrow$ we will get Symmetric)

$$\begin{cases} A \cdot A^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^T \rightarrow X = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ A^T \cdot A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^T \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \rightarrow Y = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \end{cases}$$

Symmetric

Eigen decomposition ($A = V \Lambda V^{-1}$), $V^{-1} = V^T$

$A \rightarrow U$ (left Singular vector)

$U \Rightarrow$ matrix, where cols contain Eigenvectors of $A \cdot A^T$

$A \rightarrow V$ (right Singular vector)

$V \Rightarrow$ matrix, where cols contain eigenvectors of $A^T \cdot A$

$$X = \Sigma \Sigma^T$$

$$Y = \Sigma^T \Sigma$$

Consider,

$$\Sigma = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \end{bmatrix}$$

$$X = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a^2 & 0 \\ 0 & b^2 \end{bmatrix} \Rightarrow a^2, b^2$$

Eigenvalues of $A \cdot A^T$

$$Y = \begin{bmatrix} a & 0 \\ 0 & b \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a^2 & 0 & 0 \\ 0 & b^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow a^2, b^2, 0$$

Eigenvalues of $A^T \cdot A$

$$A \cdot A^T \Rightarrow a^2, b^2 \Rightarrow \sqrt{a^2}, \sqrt{b^2} \Rightarrow (a, b) \text{ SVD} \Rightarrow \text{sqrt(Eigenvalue)}$$

$$A^T \cdot A \Rightarrow a^2, b^2 \Rightarrow \sqrt{a^2}, \sqrt{b^2} \Rightarrow (a, b) \Rightarrow \text{sqrt(Eigenvalue)}$$

$$A = U \Sigma V^T$$

Left Singular Vector ($A \cdot A^T$) (a, b)

Right Singular Vector ($A^T \cdot A$)

$\sqrt{a^2}, \sqrt{b^2}$
(Sqrt(Eigenvalues))
Singular values

i) V^T (Right Singular vectors)

ii) Σ (Singular values)

iii) U (Left Singular vectors)

$\rightarrow$ Comes from input space

$\rightarrow$ Tells which directions in i/p data are important

$\rightarrow$ In PCA, these are principal directions

$\rightarrow$ diagonal matrix

$\rightarrow$ Contains singular values (strength/importance)

$\rightarrow$ large value $\Rightarrow$ more info

$\rightarrow$ comes from o/p space

$\rightarrow$ Shows how transformed data is oriented

$\rightarrow \Sigma$ tells: - how much stretching happens in each direction.

Geometric Intuition :-

$$A = U \Sigma V^T$$

i) $V^T \rightarrow$ Rotate data

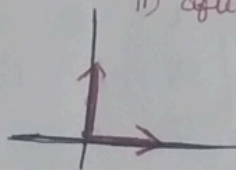
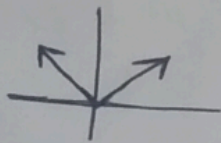
(2x3) $V \Rightarrow$ Eigenvectors of $A^T A$
Orthogonal
(3x2) (2x3) $\rightarrow$ 3x3
3D space

ii) $\Sigma = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \end{bmatrix}$ 2x3 $\rightarrow$ i/p space
O/p space

Stretch $\Sigma 2 = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ $\Sigma 1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

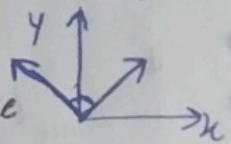
Dimension reduced from i/p 3D space to o/p 2D space

iii) U :- Rotate again

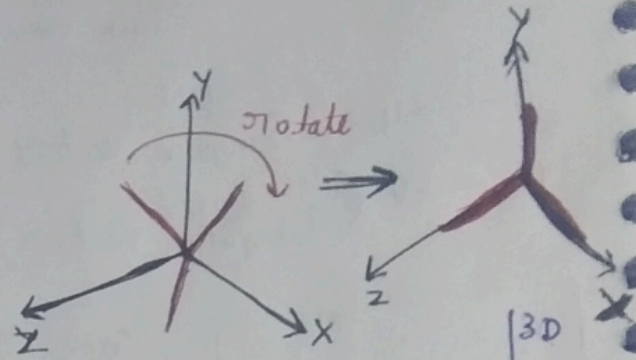


ii) after rotating U

stretching / compressing

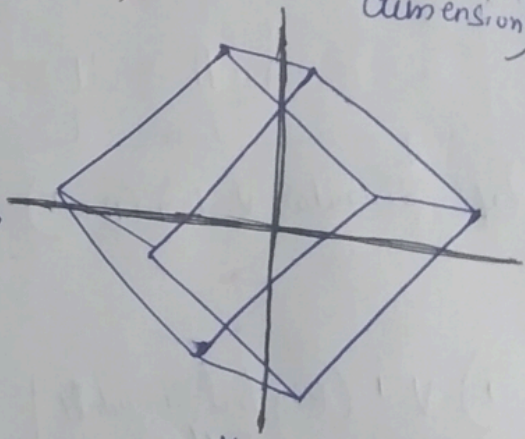
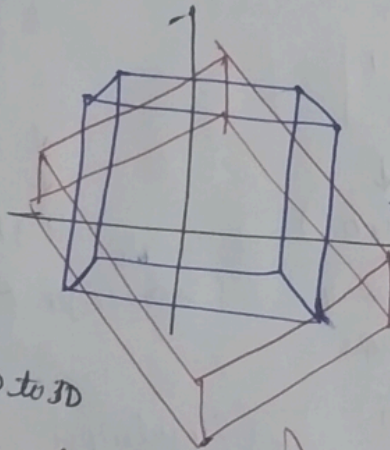
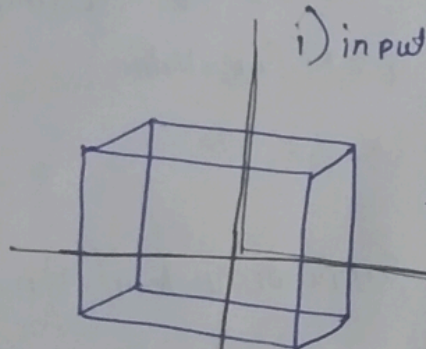


after rotating

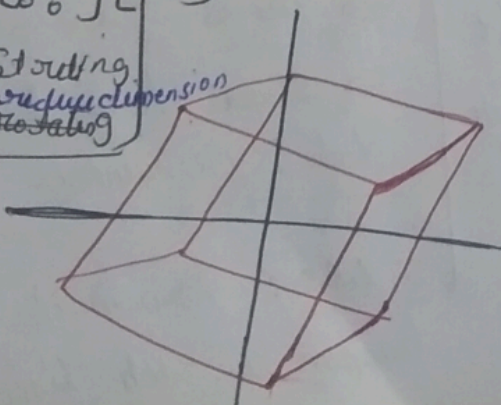
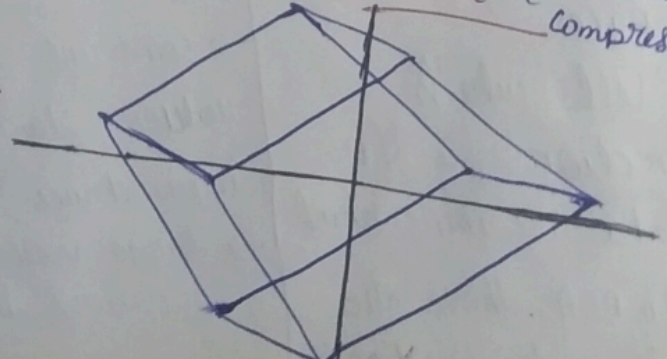


ii) V^T (rotate)

iii) $\Sigma 1$ (reduce dimension)



iv) $\Sigma 2$ (stretching / compressing)



for o/p 3x2 matrix

from 2D to 3D

$$\begin{bmatrix} a & 0 \\ 0 & b \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \Rightarrow U \text{ (rotate again)}$$

i) Storing
ii) reducing dimension
iii) rotating

Why SVD is so powerful

Because Every matrix can be decomposed, unlike eigen decomposition.

→ Eigen decomposition

⇒

works on
only Square matrices

→ SVD

⇒

Any matrix

How to calculate

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

⇒

to compute

$$U, S, V^T = \text{np.linalg.svd}(A)$$

$$\text{cov}(X) = \frac{X_c^T \cdot X_c}{n-1}$$

auto matically
in SVD

calculates this

SVD in PCA:-

→ Steps in PCA using SVD:

- Center the data
- Apply SVD
- Take top singular values and vectors
- Project data

$$\text{PCA} = \text{SVD applied to data}$$

Application of SVD:-

- Dimensionality reduction (PCA)
- Noise reduction
- Recommender system (Netflix)
- Image Compression
- Pseudoinverse computation
- NLP

Learn PCA vs LDA vs t-SNE & UMAP

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